

Durga sai prasad Pragada

Software Engineer (Python & Gen AI)

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PROFESSIONAL SUMMARY

Python & Gen AI Developer with around **2 years** of experience **building AI-powered backend applications** and **LLM-based solutions**. Proficient in **Python, FastAPI, LangChain**, and **LLM integration**, with hands-on experience developing scalable **AI pipelines using STT and TTS technologies**. Experienced in designing **high-performance REST APIs, real-time AI workflows**, and data-driven systems for production environments. Strong understanding of **backend architecture, system integration**, and **deploying reliable, scalable AI solutions**.

TECHNICAL SKILLS

Programming : Python (Core, OOP, Async)
AI / ML : LLMs (OpenAI, Groq), Prompt Engineering, NLP, Speech-to-Text (STT), Text-to-Speech (TTS) , LangChain, RAG
Backend : FastAPI, REST API Development, JWT Authentication
Databases : MySQL
Tools & Platforms : Git, GitHub, GitLab, Render, Linux

PROFESSIONAL EXPERIENCE [Deepija Telecom Private Limited, Hyderabad, India] [Aug-2024 to current]

Project 1: ConVox Platform Management System

- Developed core backend services including **Data Ingestion APIs, Operation Status APIs**, and **multi-endpoint workflow orchestration**. Designed and implemented **MySQL-based reporting modules** to monitor user activity, operational metrics, and system performance.
- Improved system performance by **~60%** using **Redis caching** for asynchronous tasks and high-frequency message routing.
- Enhanced platform stability through **structured testing, debugging, and API validation** using Postman.
- Collaborated closely with **product and engineering teams** to translate business requirements into **scalable backend solutions**.

Project 2: Real-Time AI Integration with ConVox

- Built a **Python-based WebSocket bridge** and **bidirectional server** to enable real-time communication with third-party AI services. Implemented an **event-driven architecture** to manage session initialization, live data streaming, and workflow completion events.
- Developed **low-latency processing modules** using Python **async programming and multithreading**.
- Integrated **webhook-based JSON reporting** to capture workflow results and generate detailed post-processing logs.
- Increased system throughput and reliability using **Redis caching, thread synchronization, retry mechanisms**, and **robust exception handling**.

Personal Project 1 : AI Document Assistant

- Built an AI-powered PDF summarization system that allows users to upload documents and extract key insights.
- Implemented PDF **text extraction, semantic embeddings, and vector storage** using **ChromaDB** for efficient meaning-based retrieval.
- Developed a **Retrieval-Augmented Generation (RAG)** pipeline using **Groq LLMs** to generate contextual summaries and accurate answers.
- Extended functionality to auto-generate quizzes, practice questions, and predictive question sets by analysing historical data.

Personal Project 2 : AI-Driven HR Attendance Management System [[RenderHosted](#)]

- Developed an **AI-enabled HR platform** for employee attendance tracking and workforce management.
- Implemented automated **analytics reports** to analyse **attendance trends and HR metrics**.
- Built an **AI assistant to answer HR-related queries** such as leave policies and portal usage.
- Ensured data **privacy controls** by restricting AI access to **sensitive employee information**.

INTERNSHIP EXPERIENCE [Deepija Telecom Private Limited, Hyderabad, India | Aug 2024 – Jan 2025]

- Developed and maintained data reporting modules, processing large-scale application data using **MySQL**.
- Designed and optimized database queries for efficient data retrieval and performance improvement.
- Built and tested backend APIs using **Python** and **FastAPI** for data processing and workflow execution.
- Worked on system-level integrations, improving reliability and handling of real-time data workflows.
- Progressed into **AI-based system integration**, connecting backend services with **LLM-driven** applications for intelligent automation.

EDUCATION

Masters of Computer Applications (MCA) [2021 – 2023] | CGPA: 8.0 / 10 | B V Raju College – Bhimavaram, Andhra Pradesh, INDIA.

STRENGTHS

- Strong problem-solving skills
- Clear understanding of AI workflows
- Quick learner in new AI tools
- Good communication & teamwork